WEIQI CHI

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EDUCATION

THE UNIVERSITY OF TOKYO (QS #39, 2027) TOKYO, JAPAN

Ph.D. student in Information Engineering Apr. 2024 – Mar.2027

** SPRING GX scholarship project student / Young Researcher's Encouragement Award, IEEE VTS 2024*

TOKYO INSTITUTE OF TECHNOLOGY (QS #97, 2027) TOKYO, JAPAN

Master's degree in Communication Engineering Oct. 2020 – Jun. 2022

** Supper Smart Society (SSS) scholarship project student*

RESEARCH

Learning-Based User Association & Mobility Management in mmWave V2X Networks Current

Doctoral Research

Investigated contextual multi-armed bandit (CMAB) approaches for user association (UA) in mmWave vehicular networks under non-stationary channel conditions without CSI acquisition or offline training:

- Developed BAND algorithm by integrating change-point detection (CUSUM-CD) with blockage-aware reward estimation to overcome the detection failure of conventional bandits under non-stationary channel reward brought by vehicle mobility and transient blockages. *(IEEE GLOBECOM, 2026)*
- Proposed a novel trajectory-aligned knowledge (TAK) regions in a semi-distributed CMAB framework for cross-vehicle knowledge sharing and algorithm cold-start acceleration with rigorous regret-bound analysis. *(IEEE TVT)*
- Proposed DCC-SMAB algorithm using dynamic hierarchical context clustering to mitigate slow convergence in high-dimensional context spaces for learning efficiency promotion. *(IEEE VNC, 2026)*
- Developed simulation framework integrating SUMO traffic simulator and ray-tracing tools with real-world maps for realistic V2X communication data analysis *(IEICE RCS, 2025)*.
- Proposed a geometric-based dynamic V2I blockage modeling and detection method. *(IEEE VTC, 2024)*

A Blockage-Aware mmWave V2X Connectivity Analysis based on MCM Nov. 2024 – Feb. 2025

Visiting Researcher at CARISSMA, Technische Hochschule Ingolstadt, Germany

- Collaborated with Artery core development team to design and implement ETSI ITS-G5 standards-compliant blockage prediction system with cooperative awareness on Artery simulation platform.
- Developed Maneuver Coordination Message (MCM) based V2X network simulation evaluated with OMNeT++.

Measurement and Prediction of Vegetation Attenuation on mmWave channel Sep. 2020 – Jul. 2022

Master Research & Joint Research with Rakuten Mobile, Inc.

- Designed and conducted outdoor radio measurement campaigns for static shadowing effects from vegetation to 5G base stations, validated results through ray-tracing simulation with propagation models (DKED, ITU-R).
- Presented research findings at Rakuten Mobile group discussions and student conferences (Best Oral Presenter).

FUNDINGS

ACT-X, Japan Science and Technology Agency (JST) Oct. 2026 – Mar. 2028

Passed first-round screening and interview.

ACT-X is one of the most prestigious grants available to PhD students in Japan, awarded by the country's leading national science funding body **JST** (comparable to NSF/DFG). Only 20 researchers within specified field were funded nationwide last year with direct funding of **¥6,000,000** (approx. **€33,000**) over 2 years.

PUBLICATION

Journal

[1] **Weiqi Chi**, B. Qian, H. Wu, Haibo Zhou, Manabu Tsukada, "AURA: An Automation-Level-Aware User Association and Resource Allocation Framework for AI-Native 6G V2X Networks," IEEE Communications Magazine. *(To be submit, **JCR Q1 ・IF 8.3**)*

[2] **Weiqi Chi**, B. Qian, H. Wu, D. Li, Haibo Zhou, Manabu Tsukada, "Toward Blockage-Resilient 6G-V2X Connectivity: Semi-Distributed Bandit with Dynamic Arm Set for mmWave HetNets," submitted to IEEE

Transactions on Vehicular Technology (**T-VT**), 2026. *(Under Review, **JCR Q1 ・IF 7.5+**)*

Internation conference

[1] H. Wu, B. Qian, H. Si, **Weiqi Chi**, E. Javanmardi, Haibo Zhou, Manabu Tsukada, " Sparsity and Evidential Uncertainty Enabled Communication and Sampling for Collaborative Semantic Occupancy Prediction," Proceedings of the AAAI Conference on Artificial Intelligence, 2027. *(Under Review, **CORE A***)*
[2] **Weiqi Chi**, Manabu Tsukada, "Blockage-Aware Non-stationary Dynamic Bandit for User Association in mmWave V2X Networks," IEEE Global Communications Conference (**GLOBECOM**) 2026. *(accepted, **CORE B**)*
[3] **Weiqi Chi**, X. He, and Manabu Tsukada, "Fast-Adapting V2X Networks: A Dynamic Context Clustering Multi-Armed Bandits," IEEE Vehicular Networking Conference (**VNC**), Canada, Jun. 2026.
[4] **Weiqi Chi**, J. Nakazato, T. Murakami and Manabu Tsukada, "V2I Blockage Modeling and Performance Evaluation for Connected Autonomous Vehicle", The IEEE 99th Vehicular Technology Conference (**VTC2024-Spring**), Singapore, Jun. 2024. ***(VTS Flagship)***
[5] **Weiqi Chi**, N. Keerativoranan and Jun-ichi Takada, "Outdoor mmWave Band Tx Localization Using Scattering Path under Spectrum Sharing Scenario," 13^{th} Asia-Oceania Top University League on Engineering (**AOTULE**) Conference, Daejeon, Korea, Nov. 2021.

Domestic conference

[1] **Weiqi Chi**, N. Keerativoranan, Jun-ichi Takada, "Outdoor Measurement of Millimeter-Wave Propagating Through Vegetation at 28GHz Band," MISW Conference, Tokyo, Japan, Aug. 2022. *(Chairperson)*
[2] **Weiqi Chi**, J. Nakazato, T. Murakami, Manabu Tsukada, "ミリ波帯における遮蔽対応型 V2I 通信と解決策について", in IEICE 無線通信システム研究会, Kyoto Institute of Technology, Mar. 2025.

Poster

[1] **Weiqi Chi** and M. Tsukada, "Bridging the Gap: A Realistic V2X Simulation for End-to-End Autonomous Driving," in Cooperative Interactive Vehicles (CIV) Summer School 2026, Japan, Tokyo, 2026.
[2] **Weiqi Chi** and M. Tsukada, "Blockage-aware Multi-armed Bandits for Proactive Handover in mmWave Vehicular Networks," in Cooperative Interactive Vehicles (CIV) Summer School 2025, Geneva, Switzerland, 2025.

OTHER EXPERIENCE

IEEE INFOCOM 2026, Tokyo, Japan May. 2026

Student Volunteer | Served as cameraman and logistics support staff for 900+ attendees from 40 countries

Visban Inc., Tokyo, Japan Jun. 2026 – Current

Intern | mmWave repeater simulation engineer | Sionna Raytracing, Network Controlled Repeater

Tier IV Inc., Tokyo, Japan Apr. 2026 – Sep. 2026

Intern | Autoware V2X engineer | Autoware, Veins, QuaDRiGa

Ehime Prefectural Industrial Technology Institute, Japan Aug. 2025

Intern | AR/VR Application Development | MRTK, Unity, C#, HoloLens2

General Test Systems Inc., Shenzhen, China Feb. 2023 – Dec. 2023

Intern | Application engineer | OTA testing, Anechoic chamber, VNA, Area Tester

Rakuten Mobile Inc. R&D group, drone application development, Japan Sep. 2022

Intern | Drone application development | DJI drones operation, SDK, Android platform

SKILL

- **Programming skills:** C, C++, Python, MATLAB, NED language
- **ML Frameworks & Data processing:** Scikit-learn, NumPy, Pandas
- **Development & Tools:** Git, Docker, LaTeX, Linux/Ubuntu
- **Hardware:** Vector Network Analyzer (VNA), Spectrum Analyzer, OTA testing
- **Simulation & Modeling:**
  - *VANET & V2X*: OMNeT++, SUMO, Artery, Veins
  - *Propagation*: Wireless InSite, Sionna (ray tracing), Unity, Blender (3D modeling)
- **Language:** Chinese (Native), English (Native level) , Japanese (JLPT N1) , German (Beginner)
- **Interests:** Piano, Music, Swimming, Boxing, Bouldering